

Amandeep

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EDUCATION

National Institute of Technology, Jalandhar

Bachelor of Technology

Jalandhar, India

Aug 2023 – Aug 2027

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C++, Java

AI / ML / GenAI: LangGraph, LangChain, RAG, Agentic AI, Prompt Engineering, Embeddings, Vector Databases (Qdrant), Knowledge Graphs, Cross-Encoder Reranking, PyTorch, Hugging Face, LLM Evaluation (Ragas), LightGBM, SHAP, Scikit-learn

Backend & Web: FastAPI, Node.js, Express.js, React.js, Next.js, MongoDB, REST APIs, Socket.io

DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions (CI/CD), Linux

Core CS: Data Structures & Algorithms, OOP, DBMS, Multithreading, System Design

EXPERIENCE

GiniVibe (Pre-incorporation Venture)

Aug 2026 – Oct 2026

Founding Full-Stack Engineer (Contract)

Remote

- **End-to-End Product Delivery:** Architected and launched a **cross-platform social ecosystem** from scratch as sole contracted engineer, delivering unified feature parity across **Next.js 16** (Web) and **React Native / Expo 57** (iOS & Android).
- **Service-Oriented Backend:** Engineered a **TypeScript/Node.js monorepo** with **7 microservices** and **PostgreSQL** (**Prisma ORM**), powering auth, multimedia feeds, and duplicate-protected online/offline event ticketing.
- **Real-Time Rooms & Messaging:** Built low-latency **WebRTC peer-to-peer audio/video group rooms** and 1:1 encrypted messaging with live presence detection via **WebSockets** and **Socket.io**.
- **AI Persona & Semantic Matching:** Developed an interactive **AI character chat engine** and a multi-dimensional recommendation pipeline mapping user traits and natural-language intent into ranked profile matches.
- **Cloud & Media Infrastructure:** Containerized all services using **Docker** and deployed the production ecosystem to **Microsoft Azure**, integrating **Azure Blob Storage** for high-throughput media uploads and caching.

Full-Stack Web Developer Intern

Jun 2025 – Jul 2025

Skylark Express Delhi Pvt Ltd

On-site, Gurugram

- Developed an **internal logistics dashboard**, significantly improving **data visibility** and **workflow efficiency**.
- Architected the frontend using **React**, **Next.js**, and **TypeScript**, seamlessly integrating **REST APIs**.
- Optimized **UI responsiveness** and enhanced overall **system performance** for core internal operations.

PROJECTS

OpsAgent: Autonomous SRE Fabric with End-to-End Incident Diagnosis

[\[GitHub\]](#)

Agentic AI · SRE Automation · MCP Systems · Incident Response

- Built an **autonomous AI SRE agent** for production alerts (OOM, pod crashes, 5xx) using a **7-node LangGraph workflow** + **Llama 3.3 70B**, correlating **Kubernetes** telemetry, metrics, and Git history for instant RCA.
- Architected a **RabbitMQ alert pipeline** across a **9-service FastAPI** microservices stack with **4 scoped MCP servers** (K8s, Prometheus, GitHub, Jira) and short-lived JWT auth, auto-filing Jira tickets with **Prometheus/Grafana** and **Langfuse** tracing.
- Engineered an **LLM evaluation CI pipeline** gating PRs on quality, achieving **0.968 aggregate score** across **Faithfulness**, **Answer Relevancy**, and **Context Precision** with **44 unit tests** and **10 mock incident scenarios**.

Agentic GraphRAG: Self-Correcting Research Q&A with Hybrid Retrieval + Verification

[\[GitHub\]](#)

Agentic AI · GraphRAG · Knowledge Graphs · Hybrid Retrieval

- Built an **agentic GraphRAG platform** for research Q&A fusing **Knowledge Graph** traversal, **Qdrant vector search**, and **BM25** retrieval with **Reciprocal Rank Fusion (RRF)** and **Cross-Encoder reranking**, returning grounded answers in **2–5s**.
- Designed a **self-correcting verification engine** with **stemming-aware claim-level grounding**, term-novelty penalties, 3-variant query expansion, and **multi-provider LLM support** (Llama 3.3 70B, GPT-4o, Claude) to eliminate hallucinations.
- Indexed **913 chunks** into a **325-node / 1,661-edge graph** via a **Dockerized FastAPI + Streamlit** stack, achieving a **92% verified rate**, **0.82 mean confidence**, **73% keyword hit rate**, and **62% pass rate** on an end-to-end evaluation benchmark.

SKYBRIDGE: Stateful Multi-Cloud Workload Migration & Continuity Control Plane

[\[GitHub\]](#)

Cloud Architecture · Distributed Systems · CDC · SRE · Go

- Built a **Go control plane** for stateful **AWS → Azure migration**, orchestrating cloud compatibility, drift, policy/approval, **Temporal workflows**, canary rollout, quiesce, ownership transfer, and audit across a **9-stage cutover state machine**.
- Engineered an end-to-end **PostgreSQL WAL → Debezium → Redpanda → CDC Applier** pipeline with **transactional deduplication**, forward-only checkpoints, reconciliation, and configurable **RPO-based lag gating**.
- Designed **fail-closed migration safety** with read-only canary evaluation, **exactly-once CAS ownership transfer**, and split-brain prevention; locally validated the platform with **354 automated tests**, a **12-scenario failure matrix**, and **14 Terraform modules**, with real-cloud execution deferred in v1.

CODING PROFILE

LeetCode: 450+ problems solved across Arrays, Trees, Graphs, DP | **Contest Rating 1700+** [\[Profile\]](#)